

```

root@MASTER:/etc/salt
root@MASTER:/etc/salt# salt '*' test.ping
CLIENT_1:
True
CLIENT_2:
True
root@MASTER:/etc/salt#

```

Figure 5: Salt Master-Salt Minions ping test

After the configuration of Salt Master, we can start it with the following command:

```
$sudo service salt-master start
```

After installing *Salt Minion* in the client machines, we need to edit the corresponding configuration file called *minion* (which is present in the */etc/salt* directory), using any text editor (as shown in Figure 3), in order to set the master IP address by uncommenting *master*. Before this is done, it's better to stop the Salt Minion daemon running on the client machines with the following command:

```
$sudo service salt-minion stop
```

After the configuration of Salt Minion, we can start it with the following command:

```
$sudo service salt-minion start
```

Now, Salt Master and Salt Minion will authenticate each other using the RSA public key crypto system. We can check this on the master machine, as shown in Figure 4.

After this, let's test the whole environment by giving a simple ping test, as shown in Figure 5.

Executing commands on all the Minions at one time

Let us look at the case when the administrator has to install Apache2 in all the client machines. This can be completed by executing a single command on the Salt Master as shown in Figure 6, where we can also see the responses from all the client machines.

Let us look at another case where the administrator needs to enable the firewall in all the machines. This can be completed by executing a single command on the Salt Master as shown in Figure 7. The responses from all the client machines can also be seen in Figure 7.

```

root@MASTER:~# salt '*' cmd.run 'apt-get install --yes -q apache2'
CLIENT_2:
Reading package lists...
Building dependency tree...
Reading state information...
The following extra packages will be installed:
  apache2-bin apache2-data
CLIENT_1:
Reading package lists...
Building dependency tree...
Reading state information...
Suggested packages:
  apache2-doc apache2-ssl-libs apache2-ssl-cert apache2-ssl
The following NEW packages will be installed:
  apache2

```

Figure 6: Installing Apache2 in all the Minions

```

root@MASTER:~# salt '*' cmd.run "ufw status verbose"
CLIENT_2:
Status: inactive
CLIENT_1:
Status: inactive
root@MASTER:~# salt '*' cmd.run "ufw enable"
CLIENT_1:
Firewall is active and enabled on system startup
CLIENT_2:
Firewall is active and enabled on system startup

```

Figure 7: Indicating how we can check the firewall status and enabling it in all the Minions

Salt Stack is a simple and powerful tool in the hands of systems administrators. Running commands on remote systems is its core function. Salt can execute commands across thousands of systems in seconds. 

References

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